Olist E-Commerce SQL Analysis Report

15 business questions: what the data shows, why it matters, how it helps, and what to do next

Scope

Analysis of the nine cleaned Olist CSV files loaded into PostgreSQL: orders, order items, payments, reviews, products, customers, sellers, geolocation and category names. Data covers September 2016 to October 2018. All money values are in Brazilian reais (BRL).

How each question is presented

- **The answer:** the result from the SQL query, with a chart or table.
- **Why it matters:** the business problem the question relates to.
- **How it solves the problem:** what the answer lets the business do.
- **Recommendation:** concrete actions based on the answer.

Results are observed historical patterns, not proof of cause and effect. Where a query needed correcting (for example, payments counted several times for multi-item orders), the corrected figure is used and noted in the appendix.

Executive summary

BRL 15.80M Revenue on valid orders	**98,206** Valid orders	**BRL 160.86** Average order value	**4.09 / 5** Average review score
3.12% Customers who bought again	**8.1%** Delivered orders that arrived late	**0.63%** Orders canceled	

What the data tells us

- **Growth has stalled.** Revenue rose about 3.8x from Q1 2017 to Q2 2018, then stayed flat at BRL 0.98M-1.15M per month through Aug 2018 (Q1, Q2).
- **Sellers are growing faster than demand.** Active sellers rose 32% in 2018 while orders stayed flat, so orders per seller fell about 35% from the Nov 2017 peak (Q5).
- **Customers rarely come back.** 96.9% of customers bought once; repeat customers are only 5.9% of payments (Q7).
- **Late delivery is the clearest service failure.** 8.1% of orders are late and score 2.57 stars vs 4.29 for on-time orders (Q15).
- **The business is concentrated.** Five categories make up about 39% of revenue, three states hold 66.5% of customers, and credit cards carry 78% of payment value (Q3, Q14, Q11).
- **Unhappy customers are big spenders.** One-star orders average BRL 195-202 vs about BRL 156-159 for five-star orders (Q6, Q12).
- **Profit cannot be measured yet.** The database has no cost data. Freight alone is 25%+ of price in furniture and electronics (Q13).

Top five priorities

#	Priority	Evidence	Action
1	Fix delivery reliability	Late orders: 2.57 vs 4.29 stars (Q15, Q8)	Carrier SLAs, realistic delivery estimates, proactive delay messages
2	Build a retention program	Only 3.12% repeat customers (Q7, Q14)	Post-purchase offers, repeat-rate KPI, national rollout
3	Rebalance seller growth	Orders per seller down about 35% (Q5, Q9)	Focus on seller quality and activation, not raw seller count
4	Sharpen category strategy	Top-5 share 39%; big categories rated below average (Q3, Q4, Q8)	Quality audit for low-rated big categories, push high-value categories
5	Prepare for peaks and measure better	Nov 2017 spike; flat 2018; no cost data (Q1, Q2, Q13)	Peak playbook, add cost data, fix metric definitions

Reading notes: data and method

- **Valid orders** exclude statuses 'canceled' and 'unavailable' unless a question says otherwise.
- **Revenue** means customer payments recorded in the payments table. It is not net revenue after refunds, commissions or costs.
- **Corrected metrics.** Joining items to payments repeats an order's payment for every item. Category sales, AOV and rating results use payments allocated across items, so nothing is counted twice.
- **Partial months.** The data ends in Oct 2018. Sep 2018 (BRL 3.3k) and Oct 2018 (BRL 0.6k) are cut-off effects, not a collapse. Q3 2018 contains only about two full months.
- **Ratings** are customer order ratings. The dataset has no seller-specific ratings.
- **Observed, not causal.** Patterns show association. Recommendations are hypotheses to test, for example with A/B tests, not guaranteed results.
- **Category names** appear in English where translated, and in Portuguese in Q10 and Q11 where the query returned the original names.

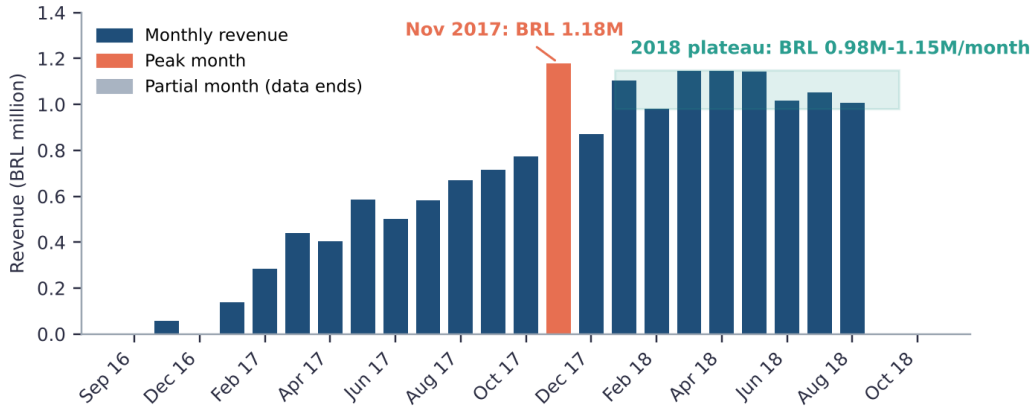
Questions covered

Q	Topic	Q	Topic
1	Revenue and trend	9	Cancellations and sellers
2	Orders and seasonality	10	Top categories by quarter
3	Popular categories	11	Payment methods
4	Average order value	12	Reviews and sales
5	Active sellers	13	Category profitability
6	Ratings and revenue	14	Customers by state
7	Repeat customers	15	Late deliveries
8	Ratings by category		

Q1 What is the total revenue generated by Olist, and how did it change over time?

THE ANSWER

BRL 15.80M from 98,206 valid orders. Revenue grew about 3.8x, then flattened in 2018.



- Quarterly revenue rose from BRL 0.86M (Q1 2017) to BRL 3.30M (Q2 2018).
- Peak month was Nov 2017 (BRL 1.18M). From Jan to Aug 2018 every month stayed between BRL 0.98M and BRL 1.15M.
- Sep and Oct 2018 look near zero because the data ends there. Q3 2018 (BRL 2.06M) is therefore partial.

WHY IT MATTERS

Revenue is the main health indicator of the marketplace. The business problem is whether growth is still happening and where the next growth will come from.

HOW IT SOLVES THE PROBLEM

The trend shows that 2017 was a ramp-up and 2018 is a plateau. More of the same (more listings, more sellers) is no longer lifting revenue, so a new growth lever is needed.

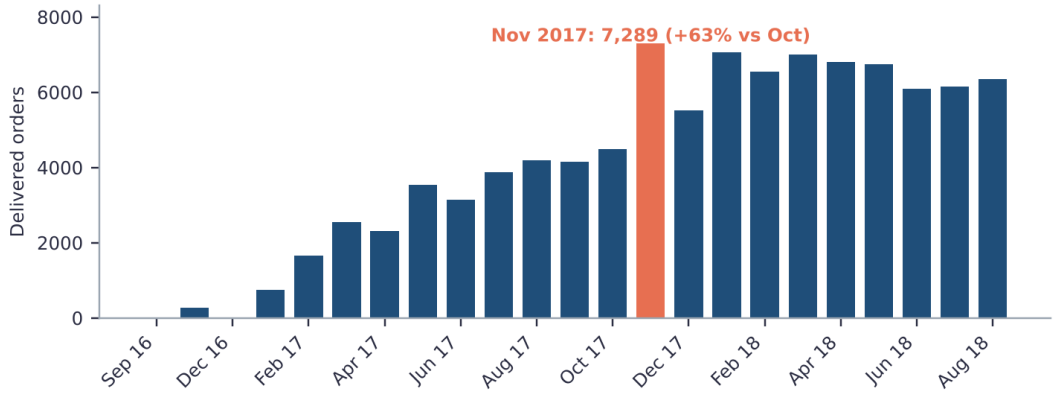
RECOM-MENDATION

1. Use the Jan-Aug 2018 level (about BRL 1.0M-1.15M a month) as the baseline for targets.
2. Move investment to levers this analysis supports: customer retention (Q7), delivery reliability (Q15) and higher-value categories (Q4).
3. Re-run the analysis with complete latest months before forecasting, and exclude Sep-Oct 2018.

Q2 How many orders were placed on Olist, and how does this vary by month or season?

THE ANSWER

98,207 valid orders. Monthly delivered orders peaked at 7,289 in Nov 2017 and have stayed between about 6,000 and 7,100 since.



- Nov 2017 was 63% above Oct 2017 (4,478 to 7,289). Dec 2017 then fell 24% to 5,513.
- 2018 ranged from 6,099 to 7,069. Jun-Aug (6,099-6,351) sat below Jan-Apr (6,555-7,069).
- Only one November exists in the data, so the spike fits a seasonal or promotional peak but cannot be confirmed as a repeating pattern.

WHY IT MATTERS

Order volume drives warehouse load, carrier capacity, seller stock and support staffing. The business problem is planning for demand surges and slowdowns.

HOW IT SOLVES THE PROBLEM

Naming the surge month and the drop that follows lets operations and marketing prepare in advance. The softer mid-2018 months are an early warning to watch.

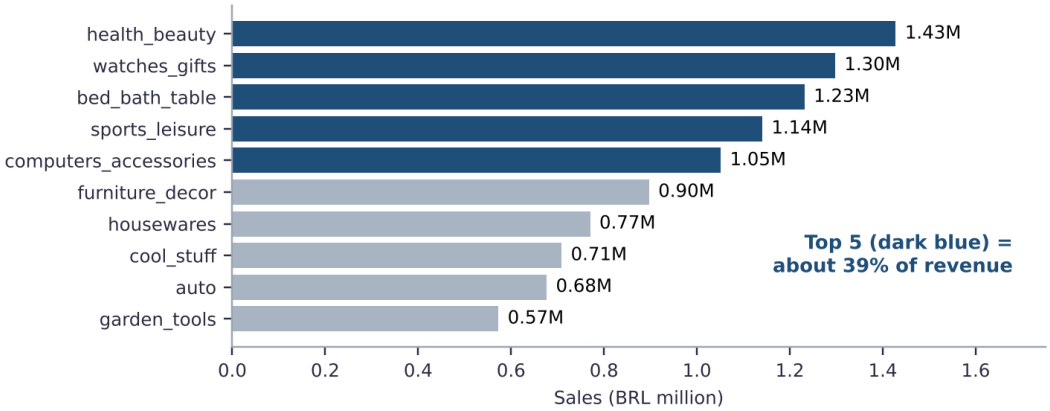
RECOM-MENDATION

1. Create a peak-season playbook (seller stock checks, carrier capacity, support staffing) that starts in October.
2. Run a re-engagement campaign in December, when orders fell 24% after the peak.
3. Track monthly orders against the 6,000-7,000 baseline and investigate if the Jun-Aug softening continues.

Q3 What are the most popular product categories, and how do their sales compare?

THE ANSWER

Health and beauty leads (BRL 1.43M), followed by watches and gifts, bed/bath/table, sports and leisure, and computer accessories.



- The top five categories total about BRL 6.15M, roughly 39% of revenue.
- Volume and value differ: bed_bath_table sells the most items (11,007) but ranks 3rd in sales. watches_gifts sells 5,943 items yet ranks 2nd (about BRL 218 per item).
- More than 30 categories each earn under BRL 50,000, a long tail of small categories.

WHY IT MATTERS

Assortment, marketing budgets and seller recruitment must be focused somewhere. The business problem is knowing which categories carry the business.

HOW IT SOLVES THE PROBLEM

The ranking shows where revenue is concentrated and which categories earn more per item, so effort goes to the categories that matter most.

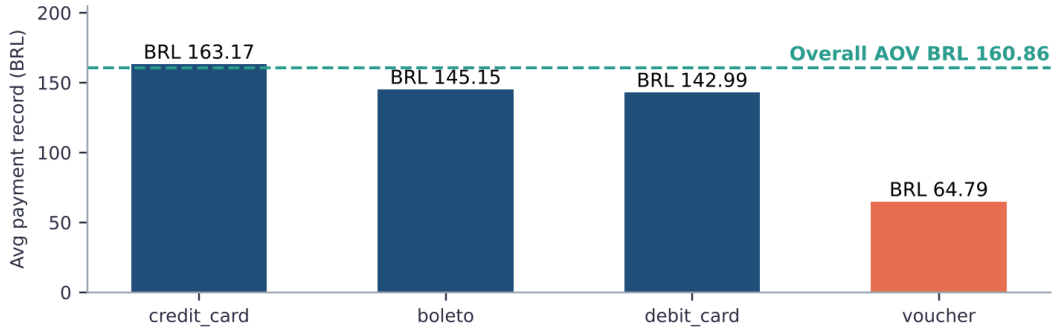
RECOM-MENDATION

1. Prioritise seller recruitment, stock depth and promotions in the top five categories.
2. Feature high-value-per-item categories such as watches_gifts and computers in prominent placements.
3. Review the long tail: keep growing only the small categories with strong ratings (Q8) and simplify navigation for the rest.

Q4 What is the average order value (AOV), and how does it vary by payment method or category?

THE ANSWER

Overall AOV is BRL 160.86 per order. Credit-card payments average BRL 163.17; voucher payments are far smaller (BRL 64.79).



- Payment records average: credit card 163.17, boleto 145.15, debit card 142.99, voucher 64.79 (BRL). Small voucher records are consistent with vouchers covering part of an order.
- By category, average payment per item ranges from BRL 1,146.80 (computers) to BRL 39.02 (home_comfort_2). This is a per-item allocation, not a conventional AOV.

Highest value per item	BRL	Lowest value per item	BRL
computers	1,146.80	home_comfort_2	39.02
small_appliances_home_oven_and_coffee	660.74	flowers	48.45
home_appliances_2	523.10	diapers_and_hygiene	54.90

WHY IT MATTERS

Revenue equals orders times AOV. With orders flat in 2018 (Q2), raising order value is one of the fastest ways to grow revenue.

HOW IT SOLVES THE PROBLEM

AOV shows the size of an average basket and which categories carry high or low tickets, so the business can design offers that lift basket size.

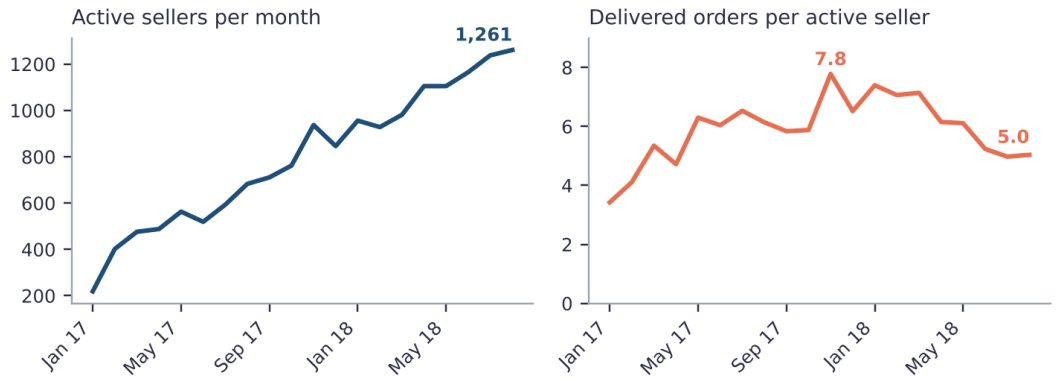
RECOM-MENDATION

1. A/B test a free-shipping or discount threshold set just above the average order, for example around BRL 180-200.
2. Cross-sell low-ticket categories (flowers, diapers, home_comfort_2) with complementary products to build larger baskets.
3. Offer card installments and financing messages on high-ticket categories such as computers and appliances.

Q5 How many sellers are active on Olist, and how does this number change over time?

THE ANSWER

Active sellers grew from 129 (2016) to 1,690 (2017) to 2,330 (2018 to Aug). Sellers kept rising in 2018 while orders stayed flat.



- Monthly active sellers rose 32% in 2018, from 956 (Jan) to 1,261 (Aug).
- Delivered orders per active seller peaked at about 7.8 in Nov 2017 and fell to about 5.0 in Aug 2018, down roughly 35%.
- Supply is growing faster than demand, so each seller gets a smaller share of orders.

WHY IT MATTERS

Sellers are the supply side of the marketplace. If sellers earn less, they leave or stop listing, which hurts assortment and service quality.

HOW IT SOLVES THE PROBLEM

The result shows the marketplace is adding sellers faster than it adds demand. That is a retention risk for sellers, not a growth plan.

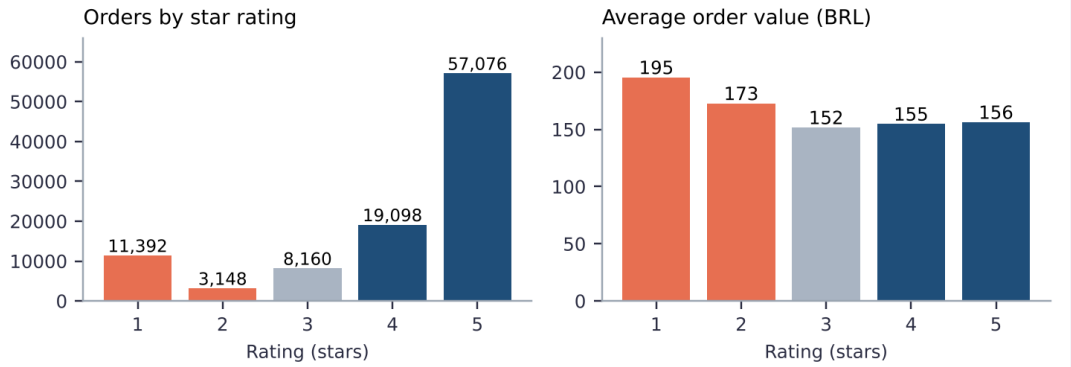
RECOM-MENDATION

1. Make orders per active seller a core KPI and track it by category and month.
2. Pause volume-based seller acquisition. Recruit only in under-supplied categories (Q3) and activate existing low-selling sellers.
3. Direct marketing spend to demand generation so the added sellers have customers to serve.

Q6 What is the distribution of ratings on Olist, and how does it affect sales performance?

THE ANSWER

77% of reviewed orders score 4-5 stars, yet ratings show no positive link to revenue (order-level correlation -0.05).



- Ratings are customer order ratings, not seller ratings. Revenue by rating: 5 stars BRL 8.96M (57,076 orders) down to 1 star BRL 2.23M (11,392 orders).
- One-star orders have the highest average value: BRL 195.38 vs BRL 156.21 for five-star orders (+25%).
- The 0.75 correlation noted in the SQL script used only five rating-group totals, which mostly reflect group size. The order-level figure (-0.05) is the reliable one.

WHY IT MATTERS

Companies often assume happier customers spend more. The business problem is whether investing in satisfaction pays back through revenue.

HOW IT SOLVES THE PROBLEM

Ratings do not predict how much a customer spends, so quality programs should not be justified by 'higher ratings mean bigger orders'. The real risk is that unhappy customers are placing larger orders.

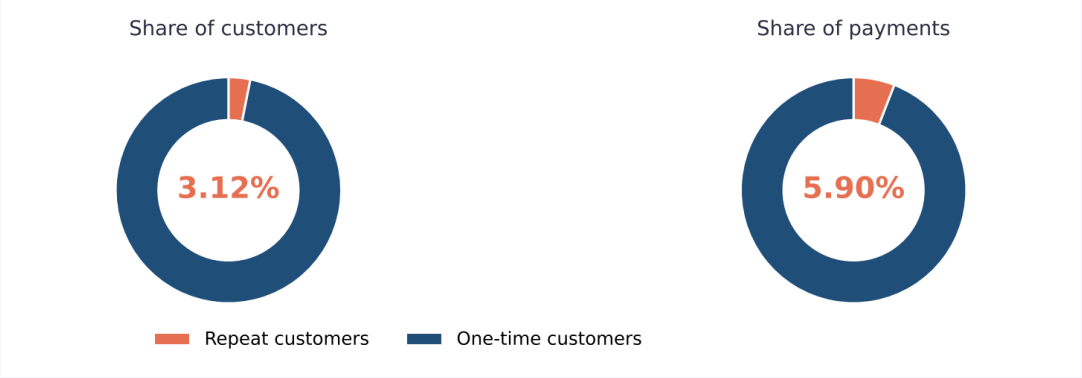
RECOM-MENDATION

1. Stop quoting the 0.75 correlation in reports and use the order-level result.
2. Give high-value orders (for example above about BRL 200) proactive tracking and priority support.
3. Read the review text of 1-2 star high-value orders to find the root causes.

Q7 How many customers have made repeat purchases, and what percentage of total sales do they account for?

THE ANSWER

Only 2,997 of 96,095 customers (3.12%) bought more than once. They account for BRL 944k, or 5.90% of payments.



- 96.9% of customers bought exactly once.
- Repeat customers spend about BRL 315 each in total.
- Illustration: each +1 percentage point in repeat rate is about 961 more repeat customers, worth roughly BRL 0.30M if they spend like today's repeaters.

WHY IT MATTERS

A marketplace that depends on first-time buyers must keep paying for new customers. The business problem is low customer loyalty with flat order growth.

HOW IT SOLVES THE PROBLEM

The result sizes the retention gap. With growth flat, even a small rise in repeat purchases is a cheaper source of revenue than acquiring new customers.

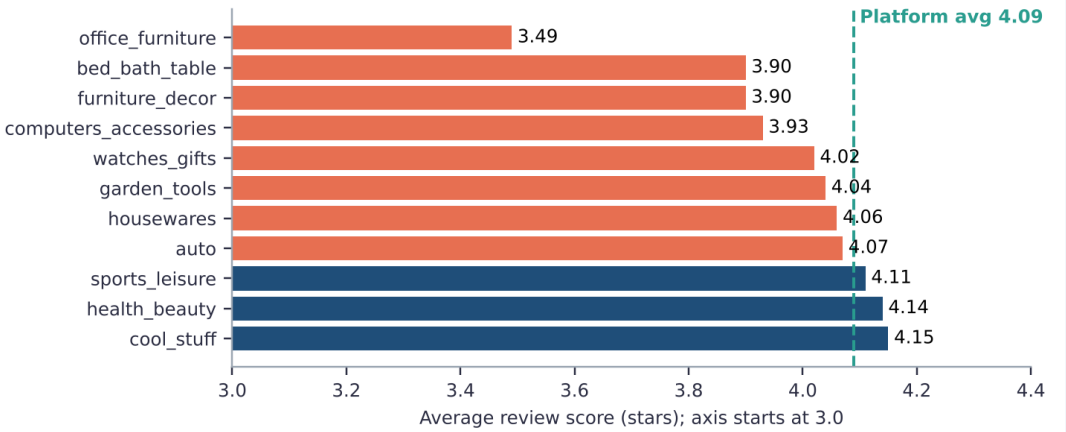
RECOM-MENDATION

1. Launch a post-purchase program with a follow-up offer 30-60 days after delivery, and test the timing.
2. Protect the first-order experience, since late first deliveries score 2.57 stars (Q15).
3. Report repeat rate and 90-day repurchase rate monthly as headline KPIs.

Q8 What is the average customer rating, and how does it vary by product category?

THE ANSWER

The average review score is 4.09 out of 5. Several large categories sit below it.



- Below average: office_furniture 3.49, bed_bath_table 3.90, furniture_decor 3.90, computers_accessories 3.93. Bed_bath_table is also a top-3 revenue category.
- Well-rated large categories include cool_stuff 4.15, perfumery 4.16, toys 4.16 and health_beauty 4.14. The best overall is books_general_interest (4.45, 508 orders).
- Very low scores in tiny categories (security_and_services 2.50 from 2 orders, kitchen_appliances 3.27 from 14) are too few orders to act on.

WHY IT MATTERS

Ratings shape customer trust and conversion. The business problem is where product or service quality is weakest, and which weak spots cost the most.

HOW IT SOLVES THE PROBLEM

The comparison shows that low ratings sit in some of the highest-revenue and bulkier categories, so quality effort can be aimed where the money is.

RECOM-MENDATION

1. Run a quality audit on office_furniture, bed_bath_table, furniture_decor and computers_accessories using review text (about 41,000 comments exist).
2. Set minimum rating standards for sellers in each category and warn sellers who fall below them.
3. Check whether late deliveries explain the low scores in these categories (see Q15).

Q9 What is the order cancellation rate, and how does it affect seller performance?

THE ANSWER

Cancellations are rare: 0.63% of orders. The seller-level list is not usable because of tiny sample sizes.

0.63%	BRL 143,256	0.9%
Orders canceled (625 of 99,441)	Payment value on canceled orders	Of valid-order revenue

- BRL 143,256 of payment value sits on canceled orders. This is not necessarily a realised loss, because no refund data is available.
- All top 20 sellers by cancellation rate show 100%, but each has only 1-2 orders. Their satisfaction scores are low (17 of the 20 average 2 stars or less), but the sample is too small to judge any seller.
- A seller with one canceled order out of one looks as bad as a seller with 500 canceled out of 500, which is why a volume threshold is needed.

WHY IT MATTERS

Cancellations break customer trust and waste logistics effort. The business problem is finding sellers who repeatedly fail to fulfil orders.

HOW IT SOLVES THE PROBLEM

The platform-wide rate is low, so a broad crackdown is not warranted. Effort should go to a small number of high-volume offenders, found with a fair ranking method.

RECOM-MENDATION

1. Rank sellers only above a minimum volume (for example 30 or more orders in a rolling 90 days; calibrate the threshold).
2. Set an alert for sellers whose cancellation rate is well above the 0.63% platform average.
3. Capture cancellation reasons and refund status so real financial loss can be measured.

Q10 What are the top-selling products on Olist, and how have their sales trends changed over time?

THE ANSWER

The query ranks categories, not individual products. Health and beauty led in 4 of 9 quarters, including the last two.

Quarter	Top category	Sales incl. freight (BRL)
Q3 2018 (partial)	health_beauty	259,950
Q2 2018	health_beauty	340,873
Q1 2018	computers_accessories	315,018
Q4 2017	watches_gifts	254,011
Q3 2017	bed_bath_table	203,104
Q2 2017	health_beauty	116,475
Q1 2017	furniture_decor	72,231
Q4 2016	furniture_decor	7,052
Q3 2016	health_beauty	143

- Five different categories led over nine quarters, so leadership shifts with the season.
- Watches and gifts led Q4 2017 (BRL 254k), consistent with holiday gifting. Q3 2018 is partial, and the 2016 quarters are very small.
- Product-level (SKU) rankings are not part of this result.

WHY IT MATTERS

Promotions, inventory and homepage space need to match what sells in each season. The business problem is knowing which categories to push and when.

HOW IT SOLVES THE PROBLEM

The quarter-by-quarter leaders show a steady traffic driver (health and beauty) and seasonal winners (gifts in Q4), which supports a promotion calendar.

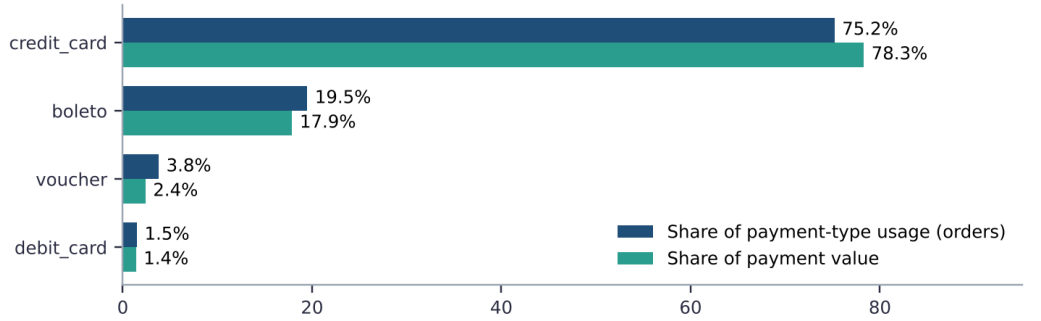
RECOM-MENDATION

1. Build a seasonal promotion calendar, with gifting categories pushed in Q4.
2. Use health_beauty as an anchor category for traffic and cross-selling.
3. Add a product-level query to find hero products inside each leading category.

Q11 Which payment methods are used most, and how does this vary by product category or region?

THE ANSWER

Credit card dominates: 76,505 orders and BRL 12.54M (78% of payment value). Boleto is second with 19,784 orders and BRL 2.87M.



- Credit card is the most used method in every state. Voucher (3,866 orders) and debit card (1,528) are small.
- Boleto is relied on more outside the big hubs: about 35-39 boleto orders per 100 credit-card orders in TO, MA, MT and RO, vs about 21 in RJ.

State	Credit-card orders	Boleto orders	Boleto per 100 credit
SP	32,074	8,205	26
RJ	10,253	2,163	21
MG	9,030	2,304	26
RS	3,967	1,359	34
MT	657	238	36
RO	184	64	35
MA	534	203	38
TO	196	76	39

WHY IT MATTERS

Payment options decide whether customers can complete a purchase. The business problem is making checkout work for every region and customer type.

HOW IT SOLVES THE PROBLEM

Knowing credit card carries most value shows where payment failures cost the most. Knowing boleto is stronger in some states shows where a second option matters.

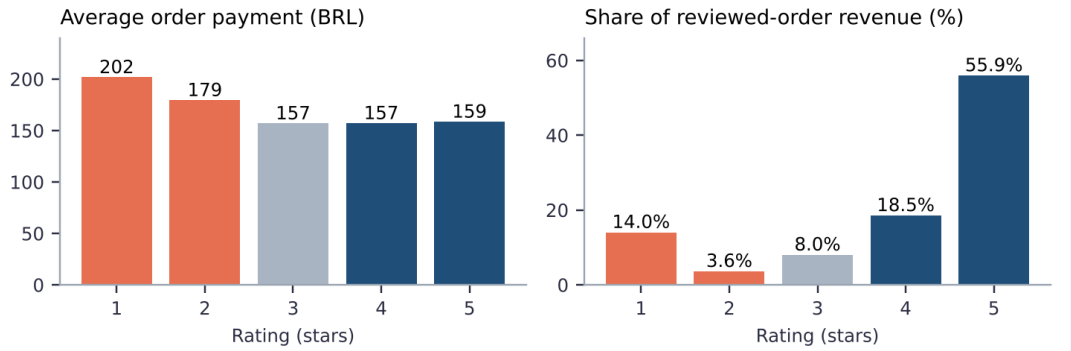
RECOM-MENDATION

1. Treat credit-card checkout (installments, fraud checks, approval rate) as business-critical and monitor the failure rate.
2. In boleto-heavy states, offer faster-settling options such as instant bank transfer (for example Pix, if not already offered) so orders are approved sooner.
3. Test targeted vouchers in regions with low card usage, and track voucher use by category.

Q12 How do customer reviews and ratings affect sales and product performance on Olist?

THE ANSWER

Orders rated 1-2 stars carry about 18% of reviewed-order revenue. One-star orders are the largest tickets (BRL 201.73 vs BRL 158.53 for five stars).



- Revenue by rating label: Excellent BRL 9.32M, Very Good 3.08M, Good 1.33M, Bad 0.60M, Very Bad 2.33M.
- One-star orders involve 7,233 different products, so poor experiences are spread across the catalogue, not a few bad products.
- Shares are indicative: an order with several reviews can appear more than once.

WHY IT MATTERS

Bad reviews damage conversion and put large revenue at risk. The business problem is protecting revenue tied to unhappy customers.

HOW IT SOLVES THE PROBLEM

Because unhappy customers place bigger orders, even a small share of bad experiences puts a large amount of money at risk, and the cause is not tied to a few products.

RECOM-MENDATION

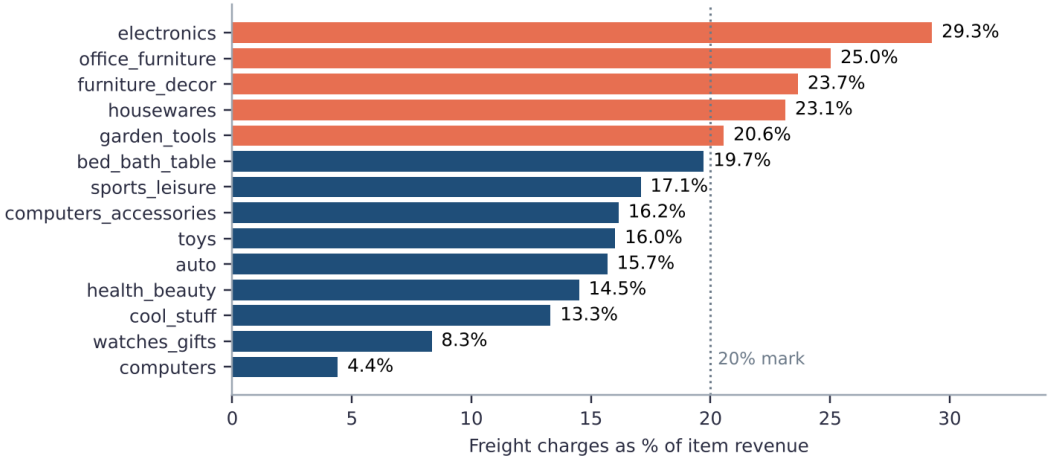
1. Create a review-recovery process: contact 1-2 star customers within 24-48 hours, with refund or replacement offers for high-ticket orders.
2. Add a 'revenue at risk' KPI (revenue on 1-2 star orders, currently about 18%) to the monthly report.
3. Look for common causes across categories, such as delivery delays (Q15), rather than product by product.

Q13 Which product categories have the highest profit margins, and how can profitability be increased?

THE ANSWER

The data cannot answer this yet. There are no product costs, commissions or operating costs. The proxy margin is about 100% for every category, so it cannot rank them.

What the data can show is the shipping burden: freight charges as a share of item revenue.



- Highest burden among large categories: electronics 29.3%, office_furniture 25.0%, furniture_decor 23.7%.
- Lowest: computers 4.4% and watches_gifts 8.3%.

WHY IT MATTERS

Profitability is the final goal, but without cost data decisions would rest on a misleading number. High shipping costs also make products less price-competitive.

HOW IT SOLVES THE PROBLEM

Freight share is a practical stand-in for logistics pressure. It flags where shipping may be hurting conversion and margins, until real costs are added.

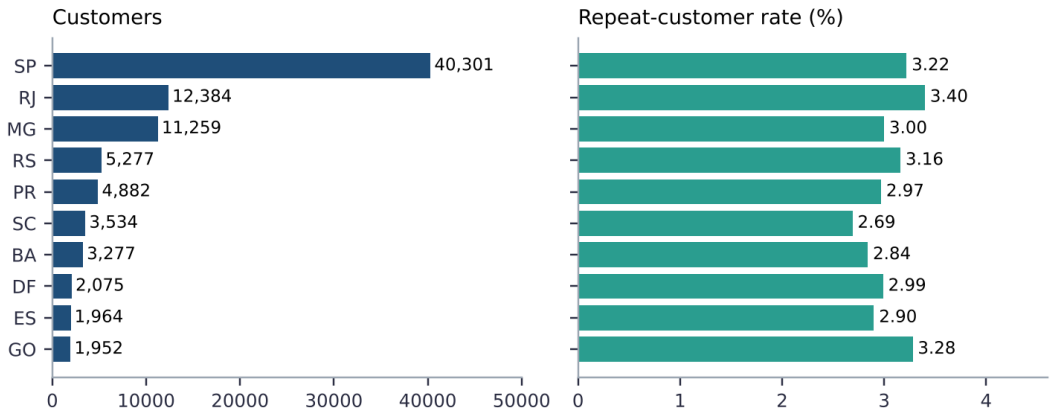
RECOM-MENDATION

1. Add product cost, marketplace commission, refund and operating cost data to the database to calculate true contribution margin.
2. For categories above 20% freight share, negotiate carrier rates, use regional fulfilment, or share shipping costs with sellers.
3. Do not use the existing proxy-profit query for decisions, since it returns about 100% for every category.

Q14 Where is customer density highest, and what is the customer retention rate by geography?

THE ANSWER

Sao Paulo, Rio de Janeiro and Minas Gerais hold 66.5% of customers (SP alone 41.9%). Repeat rate is low in every state, around 3%.



- Among states with over 1,000 customers, RJ has the highest repeat rate (3.40%). CE (1.68%) and PE (2.05%) are the lowest.
- Higher rates in AC (5.19%) and RO (4.17%) come from very small bases (77 and 240 customers).
- This uses customer state, since the query joins customers rather than map coordinates.

WHY IT MATTERS

Logistics, marketing and partnerships depend on where customers are. The business problem is spending effort where density and loyalty justify it.

HOW IT SOLVES THE PROBLEM

Density shows where service failures hit the most customers. The flat repeat rate shows retention is a national gap, not a regional one.

RECOM-MENDATION

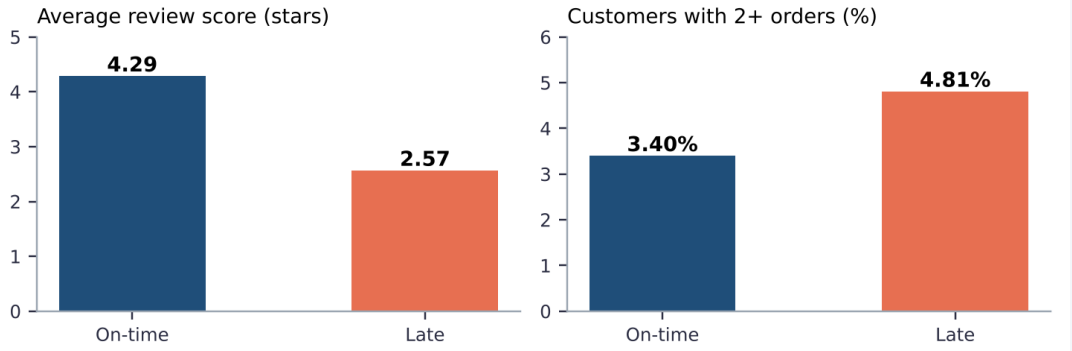
1. Protect delivery performance in the SP-RJ-MG corridor first, since it holds about two-thirds of customers.
2. Roll out the retention program nationwide, and investigate delivery times and experience in CE and PE.
3. In low-density states, test pickup points or regional fulfilment before spending on marketing.

Q15 How severely do late deliveries impact customer review scores and repurchase rates?

THE ANSWER

Late orders score 2.57 stars vs 4.29 on time, a 1.72-point (40%) drop. There is no sign that lateness lowers repurchase in this data.

8.1%	9.6 days	-1.72
Orders delivered late (7,826)	Average delay when late	Review-score drop (about 40%)



- Repeat rate is 4.81% for customers with late orders vs 3.40% for on-time customers. The measure counts any customer with 2+ orders anywhere in the data, so it cannot show cause. Treat it as inconclusive.
- The review-score gap is the clearest effect of lateness found in this analysis.

WHY IT MATTERS

Delivery is the experience customers judge most directly. The business problem is how much late delivery damages reviews and loyalty.

HOW IT SOLVES THE PROBLEM

The result puts a number on lateness: a 40% fall in satisfaction on about 1 order in 12. Cutting late deliveries is a direct way to raise ratings platform-wide.

RECOM-MENDATION

1. Agree delivery SLAs with carriers and sellers, and track the late share (8.1%) weekly with a reduction target.
2. Make delivery estimates more realistic, since 'late' is measured against the estimate, and add a buffer where history shows delays.
3. Send proactive delay messages and offer compensation for orders more than a few days late.
4. Measure true repurchase impact with a new query: first-order delivery status vs a second order within 6 months.

Consolidated action plan

Priority	Action	Questions	Metric to track
1	Reduce late deliveries and improve delivery estimates	Q15, Q8, Q12	Late share (8.1%), review score of late orders
2	Launch retention program and report repeat rate	Q7, Q14	Repeat rate (3.12%), 90-day repurchase
3	Shift seller strategy from quantity to quality and activation	Q5, Q9	Orders per active seller, cancellation rate by volume tier
4	Quality audit of low-rated large categories	Q8, Q3	Category rating vs 4.09 average
5	Lift basket size with thresholds, bundles, installments	Q4	AOV (BRL 160.86)
6	Review recovery for 1-2 star, high-value orders	Q6, Q12	Revenue at risk (about 18%)
7	Peak-season and promotion calendar	Q1, Q2, Q10	Nov and Q4 orders, post-peak drop
8	Strengthen payments for credit card and boleto-heavy states	Q11	Payment failure rate, approval time
9	Add cost data and freight-reduction measures	Q13	Contribution margin, freight share

Appendix: query issues corrected

Query	Issue in the original SQL	Treatment in this report
1	SUM(payment_value) is a payment total, not net revenue	Reported as payments on valid orders
3, 4, 6, 8, 12, 13	Joining payments to items repeats an order's payment for each item	Payments allocated across items; AOV averaged per order
4	Overall AOV averaged payment rows, not orders	Order-level AOV BRL 160.86
5	Monthly LAG ordered by month only, ignoring year	Chronological year and month order
6	Correlation across five rating groups; ratings are customer, not seller	Order-level correlation (-0.05)
7	Denominator included customers without orders; rounding before x100	Unique customers with linked orders
9	Seller join can duplicate orders in the overall rate	Rate calculated per order (0.63%)
10	Described as products but ranks categories	Reported as category ranking
13	'Profit' has no cost data behind it	Shown only as a proxy; freight share used instead